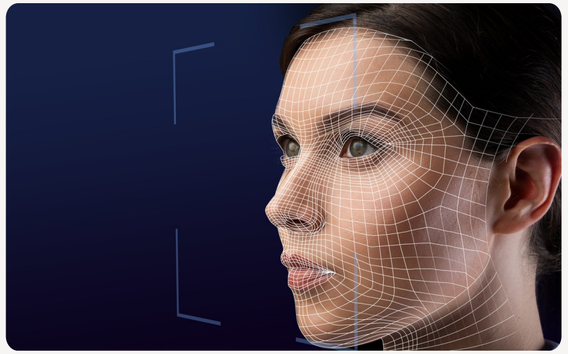
Chapter 5: The Essential You:
On the Ideas Behind AI’s New Tools

The Modern Quest to See Inside People

- In 2018, RealNetworks demonstrated a facial expression analysis program that claimed to detect gender, age, and emotions in real-time. This technology is just one example of tools being developed to “look under the hood” of people - particularly in hiring contexts.
- Companies are increasingly using AI to analyze résumés, social media activity, gameplay, facial movements, and voice patterns. All these tools share a common goal: finding a “secret formula” to predict who someone truly is and how they’ll perform.

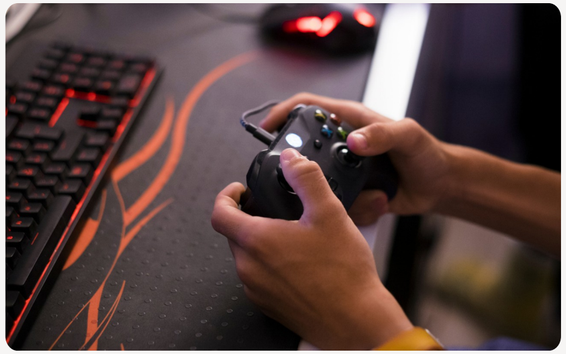


Facial Analysis

Technologies that track micro-expressions to infer emotional states.

Voice Analysis

Tools that examine tone and speech patterns to assess personality.



Gamified Assessments

Interactive tests that claim to reveal character traits through gameplay.

Digital Footprint Analysis

Systems that evaluate social media and online behavior.

The Philosophy of Essentialism

- The core assumption behind these technologies is a form of “essentialism” - the belief that humans possess stable character traits that can be categorized and compared. More importantly, it assumes we reveal our internal traits through physical appearance and digital traces.
- This idea has ancient roots in philosophical debates by thinkers like Plato and Socrates, who questioned whether our outward expressions truly reflect our inner essence. In the 18th and 19th centuries, this philosophical question transformed into a “scientific” pursuit.

Inner Essence

The belief that humans possess stable, innate character traits that form our core personality and character.

Behavioral Expression

How our supposed inner nature manifests through our actions, words, and decisions in the real world.

Physical Manifestation

The controversial idea that our appearance, expressions, and voice directly reflect our inner character traits.

AI Analysis

Modern algorithms attempting to interpret external traits to determine internal character - a technological revival of ancient philosophical questions.

Pseudoscience of the Past: Phrenology & Physiognomy

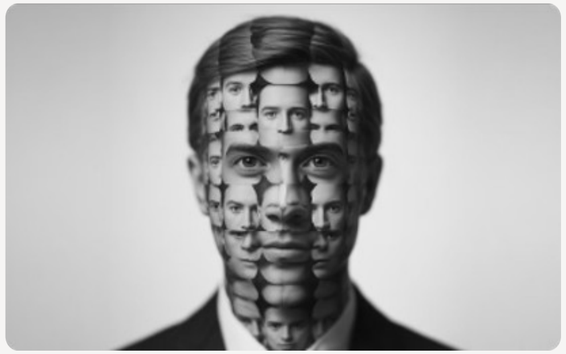
- In the 18th and 19th centuries, physiognomists and phrenologists claimed they could scientifically infer inner characteristics from physical features - just like many AI tools today. Phrenologists believed bumps on the skull revealed personality traits, while physiognomists like Cesare Lombroso claimed criminals had distinctive facial features.
- Johann Kaspar Lavater analyzed eyes, brows, mouth, and nose to determine if someone was “deceitful,” “full of malice,” or “incurably stupid.” These pseudosciences were eventually discredited but their appeal persists in new forms.

18th Century Physiognomy

Process thousands of applicants quickly without manual screening.

19th Century Phrenology

Cesare Lombroso examined criminal skulls, claiming physical traits indicated “born criminals” through distinct cranial features.



1880s Composite Photography

Francis Galton created composite photos to find “ideal typical forms” of different groups, laying groundwork for typological thinking.

21st Century AI Analysis

AI tools now claim to detect personality, criminality, and sexual orientation from facial analysis, echoing earlier pseudoscientific approaches.

Modern Iterations: AI’s Problematic Claims

Today’s AI tools often make claims eerily similar to discredited pseudosciences, attempting to find “essentialist truths” in appearance that have no scientific basis and can potentially endanger vulnerable populations.

Criminality Detection

Harrisburg University researchers claimed to have built an algorithm that could predict criminality from facial photos with “80 percent accuracy and no racial bias” - echoing Lombroso’s discredited theories about “born criminals”

Sexual Orientation Prediction

Stanford researchers built an AI “gaydar” that supposedly identified sexual orientation from photos alone, creating composite images of “archetypal” faces despite no causal relationship between facial features and orientation

Personality Assessment

AI video interviewing companies claim to evaluate traits like “adventurousness” and “resourcefulness” from 30-second recordings of facial expressions and voice

The Problem of Overfitting

Many AI tools use deep neural networks where “learning” happens in hidden layers, making them essentially black boxes. This opacity can lead to problems like “overfitting” - where algorithms identify meaningless patterns in training data rather than meaningful differences.

Black Box Learning

Deep neural networks process information in hidden layers that are difficult to interpret, creating an opaque learning process.

The Wolf vs. Husky Problem

A classic example is an AI trained to distinguish wolves from huskies that actually learned to detect snow in the background rather than the animals themselves.

Hiring Implications

Similar issues may affect facial analysis tools in hiring, where the algorithm might be responding to irrelevant features rather than job-relevant traits.

This problem occurs in a predictable cycle: algorithms are trained on data, they recognize patterns that may be coincidental, they overfit to these patterns, and ultimately make incorrect predictions when faced with new examples.

Pseudoscience in Hiring: From Graphology to AI

- Hiring managers have long been susceptible to pseudoscientific methods that promise to reveal candidates’ true nature. Graphology (handwriting analysis) was once popular in French hiring practices, with proponents claiming it revealed personality traits that couldn't be faked in interviews or résumés.

- Graphologists analyzed letter size (confidence), slant (spontaneity), and pressure (energy) - claims with no scientific evidence. Psychology professor Laurent Bègue noted, “If you ask a group of graphologists to study the same handwriting, they all come out with different interpretations.”

- **Graphology:** Claimed to reveal personality through handwriting analysis, popular in France for hiring decisions despite lack of scientific evidence.

The Ethical Implications and Path Forward

The history of physiognomy and phrenology reveals how these “sciences” were used to justify racism, classism, and discrimination. They provided seemingly objective rationales for social hierarchies and prejudice. Today’s AI tools risk repeating these mistakes if we don't critically examine their assumptions.

Recognize Historical Patterns

Understand how today’s AI assessment tools echo discredited pseudosciences of the past.

Question Essentialist Assumptions

Challenge the idea that stable personality traits can be reliably detected from external features.

Demand Scientific Evidence

Require rigorous validation that AI assessments actually measure job-relevant skills, not meaningless correlations

Prioritize Transparency

Insist on explainable AI systems where decisions can be understood and challenged.